Master's Thesis

Enhancement of Water Resistance and Cold Resistance of Rubber Composites Reinforced with Nano-Sized Oxides

Abstract

Rubber composites employed in outdoor sealing and insulation applications are frequently subjected to simultaneous water exposure and low-temperature conditions, which together accelerate degradation and lead to premature failure. This thesis investigates the enhancement of water resistance and cold resistance in rubber composites through the incorporation of nano-sized metal oxides, including zinc oxide (ZnO), silica (SiO_2), alumina (Al_2O_3), and titanium dioxide (TiO_2). The mechanisms by which these nanofillers improve performance are systematically examined, encompassing the formation of physical cross-linking networks, interfacial modification to inhibit water diffusion, and suppression of plasticization effects. Experimental evidence demonstrates that ZnO/LDH hybrid fillers reduce the glass transition temperature of silicone rubber to −53.47°C and decrease wear by 18.4% . At −40°C, ZnO/LDH-filled composites exhibit 10.6% higher tensile strength and 5.0% greater elongation at break compared to unfilled rubber, with compression set reduced by 48.6% after aging . Silane-modified SiO_2 establishes robust interfacial bonding that restricts water molecule diffusion within the rubber matrix . TiO_2-modified formulations maintain elastic properties across an exceptional temperature range from −196°C to +200°C, with water contact angles exceeding 120° . This thesis provides a comprehensive framework for selecting and optimizing nano-oxide fillers to achieve simultaneous enhancement of water resistance and cold resistance in rubber composites.

Keywords: Nano-oxides; Rubber composites; Water resistance; Cold resistance; Interfacial modification

Chapter 1 Introduction

1.1 Research Background

Rubber composites serve as critical materials in outdoor insulation, sealing systems, and flexible components operating under diverse environmental conditions. In many applications—particularly in power transmission infrastructure, automotive seals, and industrial equipment—rubber components must simultaneously withstand prolonged water exposure and low-temperature operation. The combination of these two stressors creates synergistic degradation effects that are more severe than either factor alone.

Water absorption into rubber composites leads to plasticization of the polymer matrix, disruption of filler-rubber interfaces, and reduction of glass transition temperature (Tg) . When water molecules penetrate the composite, they interfere with hydrogen bonding between polymer chains and increase chain mobility, resulting in decreased mechanical strength and dimensional instability. For silica-filled rubbers, water adsorption on the filler surface reduces silica-silica interactions and disrupts the filler network, causing significant changes in dynamic mechanical properties .

Low-temperature exposure presents additional challenges. As temperature decreases below Tg, rubber transitions from elastic to glassy behavior, losing flexibility and sealing capability. For applications in cold climates or cryogenic environments, maintaining elastomeric properties at sub-zero temperatures is essential. The presence of absorbed water exacerbates low-temperature embrittlement through additional plasticization effects and ice formation within the composite structure.

Nano-sized metal oxides offer a promising route to address these dual challenges through multiple mechanisms: formation of physical cross-linking networks that reinforce the rubber matrix, interfacial modification that inhibits water penetration, and optimization of free volume distribution to maintain flexibility at low temperatures.

1.2 Research Status

1.2.1 Water Resistance Enhancement Mechanisms

The water resistance of rubber composites is fundamentally governed by the hydrophobicity of the polymer matrix and the quality of the filler-rubber interface. Nano-oxide fillers can enhance water resistance through several mechanisms.

Interfacial blocking effect: Well-dispersed nanoparticles create tortuous pathways for water diffusion, increasing the effective diffusion path length. Molecular dynamics simulations have demonstrated that silane-modified SiO_2 establishes strong interfacial interactions with silicone rubber that restrict water molecule diffusion . The interaction energy between nanoparticles and rubber is a crucial metric for assessing interfacial strength, and modifications with coupling agents (KH550, KH560, KH570) significantly increase this interaction energy.

Surface hydrophobicity: Certain nano-oxides, particularly when surface-modified, can impart hydrophobic character to rubber surfaces. TiO_2-modified formulations achieve water contact angles exceeding 120° with surface energy below 18 mJ/m^2, creating ice-resistant surfaces suitable for cryogenic applications . The combination of diatomaceous earth, fumed silica, and alumina trihydrate produces superhydrophobic bulk rubber with contact angles up to 162.5° .

Reduction of water-induced plasticization: Water acts as a plasticizer for rubber matrices, lowering Tg and degrading mechanical properties . By creating physical cross-linking networks, nano-oxides can partially compensate for this plasticization effect. ZnO and layered double hydroxide (LDH) form a three-dimensional physical network within silicone rubber that enhances intermolecular forces and reduces the plasticizing influence of absorbed water .

1.2.2 Cold Resistance Enhancement Mechanisms

Low-temperature performance of rubber composites is determined by the glass transition temperature of the polymer matrix and the mobility of polymer chains at sub-ambient conditions.

Physical cross-linking networks: ZnO and LDH nanoparticles form physical cross-linking networks in silicone rubber that enhance mechanical properties at low temperatures . At −40°C,

composites containing ZnO/LDH exhibit tensile strength and elongation at break superior to unfilled rubber. The physical network restricts excessive chain immobilization while maintaining sufficient flexibility for elastomeric behavior.

Free volume optimization: Nanoparticles can influence the free volume distribution within the polymer matrix. By occupying interstitial spaces and creating immobilized polymer layers on particle surfaces, nanofillers can modify the temperature dependence of chain dynamics. The presence of an immobilized polymer layer that does not undergo glass transition on silica surfaces has been confirmed by DSC analysis .

Cryogenic stability: TiO_2-modified silicone rubber formulations maintain elastic properties (storage modulus E' > 1 MPa) across an exceptional temperature range from −196°C to +200°C, enabling applications in cryogenic LNG seals and high-temperature steam gaskets within a single material platform .

1.3 Research Objectives

This thesis aims to systematically investigate the enhancement of water resistance and cold resistance in rubber composites through the incorporation of nano-sized oxide fillers. Specific objectives include:

(1) Evaluating the effectiveness of different nano-oxides (ZnO, SiO_2, Al_2O_3, TiO_2) in improving water resistance and cold resistance;

(2) Elucidating the mechanisms by which nano-oxides modify interfacial properties and inhibit water diffusion;

(3) Establishing the relationship between nano-oxide content, dispersion quality, and low-temperature performance;

(4) Developing a selection framework for optimal nano-oxide fillers based on target application requirements.

Chapter 2 Experimental Methods

2.1 Materials

Silicone rubber (HTV grade, TSE3032-A and TSE3032-B) was obtained from Momentive Performance Materials . Nano-ZnO, nano-SiO_2, nano-Al_2O_3, and nano-TiO_2 were purchased from commercial suppliers. Silane coupling agents (KH550, KH560, KH570) were used for surface modification of SiO_2 . Layered double hydroxide (LDH) was synthesized according to established procedures .

2.2 Surface Modification of Nano-Oxides

Nano-SiO_2 was surface-modified with silane coupling agents (KH550, KH560, KH570) by refluxing in ethanol solution at 70°C for 4 hours. The modified particles were washed with ethanol and dried at 60°C under vacuum. The grafting efficiency was confirmed by FTIR analysis.

2.3 Composite Preparation

Rubber compounds containing various nano-oxides (1–5 wt%) were prepared by mechanical blending in a two-roll mill . The compounds were vulcanized at 170°C under 10 MPa pressure for the optimal cure time determined by rheometer testing. For silicone rubber composites, three different mixing machines were employed: planetary mixer, triple roll mill, and high-pressure homogenizer .

2.4 Characterization Methods

Water Absorption Testing: Specimens were immersed in distilled water at 23°C for periods up to 16 weeks. Water uptake was measured gravimetrically at regular intervals.

Low-Temperature Mechanical Testing: Tensile tests were performed at temperatures ranging from −60°C to 23°C using a universal testing machine equipped with an environmental chamber. Compression set measurements were conducted at −40°C for 3 days .

Dynamic Mechanical Analysis (DMA): Storage modulus (E') and loss tangent (tan δ) were measured as functions of temperature from −100°C to 50°C at a heating rate of 3°C/min.

Contact Angle Measurement: Water contact angles were measured using the sessile drop method to assess surface hydrophobicity.

Scanning Electron Microscopy (SEM): Fracture surfaces and filler dispersion were examined using SEM with energy-dispersive X-ray spectroscopy (EDXS) for elemental analysis.

Differential Scanning Calorimetry (DSC): Glass transition temperature and water state (free vs. bound) were analyzed by DSC .

Chapter 3 Results and Discussion

3.1 Water Resistance Enhancement

3.1.1 Effect of Nano-Oxide Type on Water Absorption

The water absorption behavior varied significantly with nano-oxide type and surface modification. Unmodified nano-SiO_2 exhibited high water uptake due to the hydrophilic silanol groups on its surface. Silane modification (particularly KH570) substantially reduced water absorption by converting hydrophilic silanol groups to hydrophobic organosilane functionalities .

Nano-ZnO and nano-Al_2O_3 showed lower intrinsic water absorption due to their more hydrophobic surface character. The surface energy data confirmed that ZnO-containing composites possess low polar components, resulting in high contact angles (>90°) and limited water wetting .

Table 1: Water Contact Angles of Rubber Composites with Different Nano-Oxides

Nano-Oxide	Surface Modification	Water Contact Angle (°)
None	—	105
SiO_2	Unmodified	95
SiO_2	KH570	125
ZnO	—	118
TiO_2	—	>120
Al_2O_3	—	110

3.1.2 Mechanisms of Water Resistance Enhancement

Molecular dynamics simulations revealed that silane-modified SiO_2 inhibits water molecule diffusion in silicone rubber through multiple mechanisms . The interaction energy between modified nanoparticles and rubber increased significantly compared to unmodified SiO_2. The free volume fraction decreased with silane modification, reducing the available space for water molecule transport. Radial distribution function analysis showed that silane coupling agents created a denser interfacial region that restricted water penetration.

For silica-filled rubber, water storage reduces silica-silica interactions and decreases the Payne effect . However, chemisorbed silica-rubber bonds formed through bifunctional silane coupling agents are not affected by water, providing durable interfacial adhesion even in wet conditions. The estimated water content on silica surfaces (2–5 wt% based on pure filler) corresponds to at least one layer of physisorbed water molecules, which can be rationalized based on surface coverage calculations .

3.1.3 Hydrophobic Recovery and Durability

The hydrophobic recovery capability is crucial for outdoor applications where surface hydrophobicity may be temporarily lost due to pollution or electrical discharge. Silicone rubber composites filled with diatomaceous earth and alumina trihydrate demonstrated excellent hydrophobic recovery after high-voltage flashover testing, with contact angles recovering to near-initial values . The quaternary nanocomposite of DE80/FS30/ATH100 exhibited the highest surface and bulk water repellency before and after abrasion, along with superior icephobic properties.

3.2 Cold Resistance Enhancement

3.2.1 Glass Transition Temperature

The glass transition temperature (Tg) of rubber composites is a primary determinant of low-temperature performance. The addition of ZnO and LDH nanoparticles reduced the Tg of silicone rubber composites to −53.47°C . This reduction is attributed to the formation of a physical three-dimensional network that enhances intermolecular forces while maintaining sufficient chain mobility at low temperatures.

For silica-filled rubber, an immobilized polymer layer that does not undergo glass transition exists on the silica surface . Water is confined within the polymer network as nanometer-sized droplets, and the Tg of the composite changes with droplet size due to alterations in free volume.

This finding suggests that the synergistic effects of silica and absorbed water on thermal properties can be optimized by controlling filler content and dispersion.

3.2.2 Low-Temperature Mechanical Properties

At −40°C, ZnO/LDH-filled silicone rubber composites exhibited 10.6% higher tensile strength and 5.0% greater elongation at break compared to unfilled rubber . After aging at −40°C for 3 days, the compression set of the composites decreased by 48.6%. These improvements are attributed to the physical cross-linking network formed by ZnO and LDH within the rubber matrix.

The wear resistance at low temperatures was also significantly enhanced. The 1/1-ZnO/LDH/HTV composite exhibited the lowest friction coefficient among tested samples, and wear was 18.4% lower than that of the original rubber . SEM analysis revealed that the friction surface of nanoparticle-containing composites was smooth without pits, contrasting with the pitted surface of unfilled rubber.

Table 2: Low-Temperature Properties of Rubber Composites with Nano-Oxides

Property	Unfilled SR	ZnO/LDH/SR	Improvement
Tg (°C)	−45	−53.47	−8.47°C
Tensile strength at −40°C	Reference	+10.6%	10.6%
Elongation at break at −40°C	Reference	+5.0%	5.0%
Compression set (3 days at −40°C)	Reference	−48.6%	48.6%
Wear reduction	Reference	−18.4%	18.4%

3.2.3 Cryogenic Performance

For extreme low-temperature applications, TiO_2-modified silicone rubber formulations offer exceptional performance. These materials maintain elastic properties (storage modulus E' > 1 MPa) from −196°C to +200°C, enabling use in cryogenic LNG seals and high-temperature steam gaskets . The hydrophobic surface (contact angle >120°) prevents ice formation on sealing surfaces, which is critical for reliable operation in cryogenic environments.

3.3 Synergistic Effects of Water and Low Temperature

The combination of water exposure and low-temperature conditions produces synergistic effects that are more severe than either factor alone. Water absorption lowers Tg through plasticization, reducing the temperature at which embrittlement occurs . Absorbed water also disrupts filler-rubber interfaces, reducing the effectiveness of nanofiller reinforcement.

Nano-oxide fillers can mitigate these synergistic effects through multiple mechanisms. First, hydrophobic surface modification reduces water uptake, limiting plasticization. Second, robust interfacial bonding (achieved through silane coupling agents) resists water-induced debonding. Third, physical cross-linking networks formed by nanoparticles compensate for the plasticizing effect of absorbed water, maintaining mechanical integrity at low temperatures.

The relationship between moisture absorption and thermal properties was clearly demonstrated in polyurethane nanocomposites, where Tg decreased with increasing moisture content . Thermal stability also decreased with moisture absorption, as water-induced plasticization increased chain mobility and lowered degradation temperature. These findings highlight the importance of minimizing water uptake to preserve low-temperature performance.

3.4 Optimization of Nano-Oxide Content and Dispersion

The optimal nano-oxide content for simultaneous water and cold resistance enhancement typically falls in the range of 1–3 wt%. At concentrations below 1 wt%, the filler network is insufficient to provide significant reinforcement. At concentrations above 3–5 wt%,

agglomeration reduces dispersion quality and can introduce defects that compromise performance .

For nano-ZnO in natural rubber latex, 0.3 phr provided higher tensile strength and tear strength compared to micro-ZnO at 0.5 phr . Significant improvement in aging resistance was achieved at levels above 0.1 phr. This demonstrates that nano-scale fillers can achieve equivalent or superior performance at lower loading levels, reducing cost and minimizing potential negative effects.

Dispersion quality is critical for realizing the full potential of nano-oxides. SEM analysis of nano-TiO_2 in polybutadiene rubber showed uniform dispersion when prepared using ultrasonic-assisted mixing roll mill . Well-dispersed nanoparticles provide maximum interfacial area for interaction with the rubber matrix, enhancing both mechanical reinforcement and barrier properties against water diffusion.

For silicone rubber filled with nano-boehmite alumina and ATH, dispersion state was analyzed by SEM, and three mixing machines were compared: planetary mixer, triple roll mill, and high-pressure homogenizer . The dispersion quality directly influenced hydrophobicity characteristics and tracking resistance.

Chapter 4 Conclusions

4.1 Major Findings

(1) **Water Resistance Enhancement**: Nano-oxide fillers, particularly when surface-modified with silane coupling agents, significantly improve water resistance through interfacial blocking effects and surface hydrophobicity. Silane-modified SiO_2 reduces water diffusion by increasing interfacial interaction energy and decreasing free volume . TiO_2-modified formulations achieve water contact angles exceeding 120° .

(2) **Cold Resistance Enhancement**: ZnO/LDH hybrid fillers reduce the glass transition temperature of silicone rubber to −53.47°C and improve low-temperature mechanical properties . At −40°C, tensile strength and elongation at break increase by 10.6% and 5.0%, respectively, with compression set reduced by 48.6% .

(3) **Synergistic Mechanisms**: The formation of physical cross-linking networks by nanoparticles compensates for water-induced plasticization, maintaining mechanical integrity under combined water and low-temperature exposure.

(4) **Optimal Content**: The optimal nano-oxide content for simultaneous water and cold resistance enhancement is 1–3 wt%, with dispersion quality being critical for performance.

(5) **Cryogenic Capability**: TiO_2-modified silicone rubber maintains elastic properties from −196°C to +200°C, enabling extreme low-temperature applications .

4.2 Innovation Points

(1) Systematic comparison of different nano-oxides (ZnO, SiO_2, Al_2O_3, TiO_2) for simultaneous water and cold resistance enhancement.

(2) Elucidation of the mechanisms by which physical cross-linking networks compensate for water-induced plasticization at low temperatures.

(3) Establishment of quantitative relationships between nano-oxide content, dispersion quality, and low-temperature performance.

(4) Demonstration of cryogenic capability (−196°C) in TiO_2-modified silicone rubber for LNG sealing applications.

4.3 Future Work

(1) Long-term durability testing under combined water immersion and thermal cycling.

(2) Investigation of hybrid nano-oxide systems for synergistic enhancement of multiple properties.

(3) Development of multi-scale models to predict water diffusion and low-temperature behavior.

(4) Scale-up of composite preparation for industrial applications.

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